

Dr. Paul Schmidt

DATA SCIENTIST / BIOSTATISTICIAN

Hamburg, Germany

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🔗 ResearchGate



Professional Experience

Data scientist / General Manager

Hamburg, Germany

BioMath - Applied Statistics and Informatics in Life Sciences

Since 2019-01

- End-to-end analytics: clarify questions, acquire/align data, build models, translate results for both technical and executive audiences
- Selected projects: time-series & regression on air parameters; agricultural treatment comparisons; monitoring design & evaluation; epidemiological meta-analyses; spatial analysis with GIS
- Scaled SOPs (e.g., for systematic reviews & meta-analyses): consolidated toolchain, added automation, improved reproducibility and cut cycle time
- Systematic reviews led; authored, edited, and finalized scientific reports/publications
- General manager since 2022

Workshop teacher

see 'Workshops' section below

Freelancer (part-time)

Since 2018-11

- Develop and teach workshops on statistics in R; specific content and language according to the contractor
- Provide corresponding material on my websites (see 'Other skills' section below)

Research associate

Stuttgart, Germany

University of Hohenheim

2015-09 - 2018-12

- Personalized consulting (ranging from single-appointment to project-accompanying) for students and research associates in terms of experimental design, data handling, statistical analyses and/or presentation of results
- Develop, conduct and manage yearly statistical analysis of yield stability data for external company
- Develop, organize and teach workshops in statistics, R and SAS
- Supervise student writing his MSc thesis

Junior Data scientist

Rostock, Germany

BioMath - Applied Statistics and Informatics in Life Sciences

2015-01 - 2015-08

- Streamline statistical analyses of monitoring data
- Implement SOP for systematic literature reviews

Education

Dr. sc. agr.

Stuttgart, Germany

University of Hohenheim

2015-09 - 2019-11

- DFG-funded PhD student in the biostatistics unit of Prof. Dr. Hans-Peter Piepho
- Cumulative doctoral thesis: 'Estimating heritability in plant breeding programs' graded 'magna cum laude'

Visiting PhD student

West Lafayette, IN, USA

Purdue University

2015-09 - 2015-12

- Visiting PhD student in the statistical bioinformatics unit of Prof. Dr. Rebecca Whitbeck Doerge
- Arranged on personal initiative, this collaboration allowed for scientific exchange and inspiration at the beginning of my PhD

MSc Crop Science: Plant Breeding

Stuttgart, Germany

University of Hohenheim

2012-10 - 2014-12

- Specialization in biostatistics and plant breeding (final grade 1.4)
- MSc Thesis: 'Statistical Evaluation and Analysis of PACTS trials as a series of on-farm strip trials without replicates' graded 1.0

BSc Agribiology (in german)

University of Hohenheim

Stuttgart, Germany

2009-10 - 2012-09

- Specialization in plant sciences and genetics (final grade 1.9)
- BSc Thesis: 'Cumulative effects of glyphosate trace concentrations during root exposition of winter wheat' graded 1.0

Student exchange year

Alexander Central High School

Taylorsville, NC, USA

2006-08 - 2007-07

- Completed senior year and obtained high school diploma

Skills

Analytics & Statistics

(Generalized) Linear/Mixed Models, Exploratory & descriptive analysis, Experimental design, Machine Learning

Communication & Impact

Executive-ready insights, Data visualization, Reports/scientific publications, Talks/workshops

General

collaboration, communication, structuring, time management, strategic oversight, problem solving

Languages

German (native), English (fluent)

Open Source

Website https://schmidtpaul.github.io/dsfair_quarto/, R package BioMathR <https://schmidtpaul.github.io/BioMathR/>, R package CitaviR <https://schmidtpaul.github.io/CitaviR/>

Software & Tools

R, Python, SAS, SPSS, LLM-assisted workflows, Quarto/Markdown, Markdown, Version Control (git), reproducible pipelines & DAG workflows, SQL, MS Office (VBA)

Workshops

2026

- 2026 Apr — Data Science with R (pt. 2) — Max Planck Inst. Tübingen via zoom

2025

- 2025 Dec — Statistics with R - an Introduction — Universität Bonn via zoom
- 2025 Nov — exp. Design - Practicals in R — CIHEAM Zaragoza via zoom
- 2025 Nov — Data Science with R - an Introduction — Max Planck Inst. Tübingen via zoom
- 2025 Oct — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2025 Jul — Statistics with R - an Introduction — Universität Bonn via zoom
- 2025 May — Statistics with R (Beginner) — Universität Kassel via zoom
- 2025 Apr — R-Statistics for Beginners — Pro-RUWA (e-learning)
- 2025 Apr — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2025 Mar — Data Scientist - Focus Python — Karrierecenter der Bundeswehr (e-learning)
- 2025 Mar — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)

- 2025 Mar — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2025 Mar — Data Science in den exp. Naturwiss. (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2025 Mar — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2025 Feb — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom

2024

- 2024 Dec — Statistics with R - an Introduction — Universität Bonn via zoom
- 2024 Oct — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Sep — Data science with R for scientists — Universität Rostock via zoom
- 2024 Sep — Statistics with R — Universität Hamburg via zoom
- 2024 Sep — R and the {Tidyverse} — TU Dortmund
- 2024 Aug — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Jul — Statistics with R - an Introduction — Universität Bonn via zoom
- 2024 Jun — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2024 Jun — Data Science in den exp. Naturwiss. (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2024 Apr — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2024 Apr — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2024 Apr — Data Science with R (pt. 2) — Max Planck Inst. Tübingen via zoom
- 2024 Apr — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Mar — Data Analytics mit Python — Karrierecenter der Bundeswehr (e-learning)
- 2024 Feb — Advanced data visualization in R — 70th Biometrical Colloquium, Lübeck

2023

- 2023 Dec — Feldversuche und Statistik - Interaktive Beratung — Hochschule Nürtingen-Geislingen via zoom
- 2023 Dec — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2023 Dec — Data Science in den exp. Naturwiss. (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2023 Dec — Statistics with R - an Introduction — Universität Bonn via zoom
- 2023 Nov — exp. Design - Practicals in R — CIHEAM Zaragoza via zoom
- 2023 Nov — Data Science with R - an Introduction — Max Planck Inst. Tübingen via zoom
- 2023 Oct — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2023 Oct — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2023 Jul — R Introduction — Universität Flensburg via zoom
- 2023 Jul — Statistics with R - an Introduction — Universität Bonn via zoom
- 2023 Jun — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2023 Jun — Data Science in den exp. Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2023 May — Statistics with R - an Introduction — Universität Bonn via zoom
- 2023 May — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2023 May — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2023 Feb — Introduction to data science for exp. life sciences with R — Pro-RUWA via zoom

2022

- 2022 Nov — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Data Science in den exp. Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2022 Nov — Statistics with R - an Introduction — Universität Bonn via zoom
- 2022 Oct — R and the {Tidyverse} — FBN, Dummerstorf via zoom
- 2022 Mar — Data science for exp. life sciences with R (pt. 2) — Forsch.Einr. BMEL via zoom
- 2022 Mar — Data Science in den exp. Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2022 Mar — Data science for exp. life sciences with R (pt. 1) — Forsch.Einr. BMEL via zoom
- 2022 Mar — Data Science in den exp. Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom

2021

- 2021 Dec — Statistics with R (Beginner) — Universität Kassel
- 2021 Jul — Data science in den Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom
- 2021 May — Data science in den Naturwiss. mit R (Tl. 1) — Forsch.Einr. BMEL via zoom
- 2021 Mar — Data science in den Naturwiss. mit R (Tl. 2) — Forsch.Einr. BMEL via zoom

2020

- 2020 Nov — Planning exp. designs, repeated meas., and their analyses in R — Universität Kassel via zoom
- 2020 Nov — Data science in den Naturwiss. mit R (Tl. 1) — Forsch.Eintr. BMEL via zoom
- 2020 Oct — exp. Design - Practicals in R — CIHEAM Zaragoza via zoom
- 2020 Mar — Real-time consultation on statistics and mixed models in R — Universität Kassel

2019

- 2019 Dec — Basics of applied statistics — Universität Rostock
- 2019 Nov — Data science in den Naturwiss. mit R (Tl. 2) — Forsch.Eintr. BMEL, Braunschweig
- 2019 Oct — Data science in den Naturwiss. mit R (Tl. 1) — Forsch.Eintr. BMEL, Braunschweig
- 2019 Sep — Essential basics of statistics — Universität Rostock

2018

- 2018 Nov — Gemischte Modelle in R — Forsch.Eintr. BMEL, Braunschweig
- 2018 May — Implementation of yield stability assessment with ASReml-R — Bangladesh Rice Res. Inst., Gazipur

2016

- 2016-2018 — Statistical analysis with SAS (monthly) — Universität Hohenheim, Stuttgart
- 2016-2018 — Statistical analysis with R (monthly) — Universität Hohenheim, Stuttgart

Publications

Buntaran, Harimurti, Hans-Peter Piepho, Paul Schmidt, Jesper Rydén, Magnus Halling, and Johannes Forkman. 2020. "Cross-Validation of Stagemixed-Model Analysis of Swedish Variety Trials with Winter Wheat and Spring Barley." *Crop Science* 60 (5): 2221–40. <https://doi.org/10.1002/csc2.20177>.

Friedrichs, Paula, Kerstin Schmidt, Paul Schmidt, Daniel Schöttle, and Kai Vogeley. 2025. "Autismus-Spektrum-Störung Und Kraftfahreignung: Schlussbericht." Edited by BAST. <https://doi.org/10.60850/fv-m2>.

Friedrichs, Paula, Paul Schmidt, and Kerstin Schmidt. 2021. "Protanopie Und Protanomalie Bei Berufskraftfahrern Und Berufskraftfahrerinnen: Prävalenz Und Unfallrisiko: Schlussbericht Zur Durchführung Einer Systematischen Literatur-/ Datenrecherche Und Datenauswertung." *Berichte Der Bundesanstalt für Straßenwesen*. <https://www.bast.de/DE/Publikationen/Berichte/unterreihe-m/2022-2021/m319.html>.

Friedrichs, Paula, Paul Schmidt, Jörg Schmidtke, Kerstin Schmidt, and Hans Hauner. n.d. "Systematische Recherche Nach Daten Zu Den Auswirkungen Der Covid-19-Pandemie Auf Die Lebenssituation Und Versorgung von Menschen Mit Diabetes Mellitus: Studienprotokoll." <https://osf.io/n25hv>.

Kukowski, Sina, Paul Schmidt, Hans-Peter Piepho, Markus Röhl, Hans-Karl Hauffe, and Thilo Streck. 2020. "Auswirkungen Atmosphärischer Stickstoffeinträge Auf Magere Flachland-mähwiesen in Baden-württemberg." *Natur Und Landschaft* 95 (2): 58–67. <https://doi.org/10.17433/2.2020.50153773.58-67>.

Rahman, Niaz Md Farhat, Waqas Ahmed Malik, Md Shahjahan Kabir, Md Azizul Baten, Md Ismail Hossain, Debi Narayan Rudra Paul, Rokib Ahmed, et al. 2023. "50 Years of Rice Breeding in Bangladesh: Genetic Yield Trends." *TAG. Theoretical and Applied Genetics. Theoretische Und Angewandte Genetik* 136 (1): 18. <https://doi.org/10.1007/s00122-023-04260-x>.

Schlenz, Anna, Jakob-Maximilian Keller, Hanna Christina Schmidt, Paul Schmidt, and Kerstin Schmidt. 2024. "Vulnerabilität älterer Menschen Gegenüber Luftverunreinigungen, Klimawandel, lärm Und Chemikalien (Literaturstudie): Forschungskennzahl 3718 62 201 0." Edited by Umweltbundesamt. Umwelt & Gesundheit. Dessau-Roßlau. **im Druck**.

- Schmidt, Kerstin, Paula Friedrichs, Hanna Cornelsen, Paul Schmidt, and Thomas Tischer. 2021. "Musculoskeletal Disorders Among Children and Young People: Prevalence, Risk Factors, Preventive Measures: A Scoping Review." EU OSHA. <https://doi.org/10.2802/511243>.
- Schmidt, Kerstin, Paula Friedrichs, and Paul Schmidt. 2022. "Warenstromanalyse Tierischer Lebensmittel: Gutachten Zur Erzeugung, Verarbeitung, Vermarktung Und Zum Verzehr von Fleisch, Milch Und Eiern in Deutschland." Umweltbundesamt. https://www.umweltbundesamt.de/sites/default/files/medien/479/publikationen/texte_158-2022_warenstromanalyse_tierischer_lebensmittel.pdf.
- Schmidt, Kerstin, Anna Schlenz, Paul Schmidt, and Dina Tovar. 2023. "Understanding of 'Negligible Exposure' in International Science, Policy and Law, and Qualification of the Term in Relation to the Exposure of Endocrine Substances to the Environment: To Support the Interpretation of 3.8.2 of Annex II of Regulation (EC) No 1107/2009." Umwelt & Gesundheit. https://www.umweltbundesamt.de/sites/default/files/medien/11850/publikationen/10_2023_uug_understanding_of_negligible_exposure.pdf.
- Schmidt, Kerstin, Paul Schmidt, Marieluise Lang, and Alexander Rusam. 2020. "Einsatz Und Potential Biobasierter Additive in Kunststoffen: Abschlussbericht." Fachagentur Nachwachsende Rohstoffe. <https://www.fnr.de/presse/pressemitteilungen/aktuelle-mitteilungen/aktuelle-nachricht/biobasierte-additive-fuer-kunststoffe>.
- Schmidt, Kerstin, Paul Schmidt, and Anna Schlenz. 2023. "Challenges for Standardized Ergonomic Assessments by Digital Human Modeling." In *Digital Human Modeling and Applications in Health, Safety, Ergonomics and Risk Management*, edited by Vincent G. Duffy, 215–41. Cham: Springer Nature Switzerland. <https://doi.org/10.1007/978-3-031-35741-1{textunderscore }18>.
- Schmidt, Kerstin, Paul Schmidt, and Thomas Tischer. 2021. "Musculoskeletal Disorders Among Children and Young People: Prevalence, Risk Factors, Preventive Measures: OSHwiki." Edited by EU OSHA. <https://oshwiki.osha.europa.eu/en/themes/musculoskeletal-disorders-among-children-and-young-people-prevalence-risk-factors-preventive>.
- Schmidt, Kerstin, Jörg Schmidtke, and Paul Schmidt. 2020. "Studie Zum Potenzial von Wirtschaftsdünger Zur Energetischen Verwertung Im Land Brandenburg (Referenz: VV-0039-2017): Schlussbericht." <https://mleuv.brandenburg.de/sixcms/media.php/9/Potenzial-Wirtschaftsduenger-energetische-Verwertung-Brandenburg.pdf>.
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- Schmidt, Kerstin, Jörg Schmidtke, Paul Schmidt, Kohl, Christian, Ralf Wilhelm, Joachim Schiemann, Hilko van der Voet, and Steinberg, Pablo. 2017. "Variability of Control Data and Relevance of Observed Group Differences in Five Oral Toxicity Studies with Genetically Modified Maize MON810 in Rats." *Archives of Toxicology* 91 (4): 1977–2006. <https://doi.org/10.1007/s00204-016-1857-x>.
- Schmidt, Paul. 2019. "Estimating Heritability in Plant Breeding Programs." Dissertation to obtain the doctoral degree of Agricultural Sciences (Dr. sc. agr.), Stuttgart: University of Hohenheim; Biostatistics unit (340c), Institute of Crop Science (340). <http://opus.uni-hohenheim.de/volltexte/2020/1720/>.
- Schmidt, Paul, Jens Hartung, Jörn Bennewitz, and Hans-Peter Piepho. 2019. "Heritability in Plant Breeding on a Genotype-Difference Basis." *Genetics* 212 (4): 991–1008. <https://doi.org/10.1534/genetics.119.302134>.
- Schmidt, Paul, Jens Hartung, Jürgen Rath, and Hans-Peter Piepho. 2019. "Estimating Broad-Sense Heritability with Unbalanced Data from Agricultural Cultivar Trials." *Crop Science* 59 (2): 525–36. <https://doi.org/10.2135/cropsci2018.06.0376>.
- Schmidt, Paul, Jens Möhring, Richard Jürgen Koch, and Hans-Peter Piepho. 2018. "More, Larger, Simpler: How Comparable Are on-Farm and on-Station Trials for Cultivar Evaluation?" *Crop Science* 58 (4): 1508–18. <https://doi.org/10.2135/cropsci2017.09.0555>.

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———. 2016. "Post Market Monitoring of Insect Protected Bt Maize MON 810 in Europe: Biometrical Annual Report on the 2015 Growing Season." https://ec.europa.eu/food/sites/food/files/plant/docs/gmo_rep-stud_mon-810_report-2015_app-01.pdf.

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Schmidt, Paul, Jörg Schmidtke, and Kerstin Schmidt. 2019. "Post Market Monitoring of Insect Protected Bt Maize MON 810 in Europe: Biometrical Annual Report on the 2018 Growing Season." https://ec.europa.eu/food/sites/food/files/plant/docs/gmo_rep-stud_mon-810_report-2018_app-01.pdf.

Schmidtke, Jörg, Paul Schmidt, and Kerstin Schmidt. 2024. "Aufarbeitung Und Analyse von Luftqualitätsdaten Der Deutschen Demokratischen Republik." <https://www.umweltbundesamt.de/publikationen/aufarbeitung-analyse-von-luftqualitaetsdaten-der>.

Tulinská, Jana, Karine Adel-Patient, Hervé Bernard, Aurélia Líšková, Miroslava Kuricová, Silvia Ilavská, Mira Horváthová, et al. 2018. "Humoral and Cellular Immune Response in Wistar Han RCC Rats Fed Two Genetically Modified Maize MON810 Varieties for 90 Days (EU 7th Framework Programme Project GRACE)." *Archives of Toxicology* 92: 2385–99. <https://doi.org/10.1007/s00204-018-2230-z>.

Zeljenková, Dagmar, Radka Aláčová, Júlia Ondrejková, Katarína Ambušová, Mária Bartušová, Anton Kebis, Jevgenij Kovrižnych, et al. 2016. "One-Year Oral Toxicity Study on a Genetically Modified Maize MON810 Variety in Wistar Han RCC Rats (EU 7th Framework Programme Project GRACE)." *Archives of Toxicology* 90 (10): 2531–62. <https://doi.org/10.1007/s00204-016-1798-4>.